

Autocorrelation Function Overview

- Definitions
- Properties
- Biased and unbiased estimators

Autocorrelation Function Defined

Normalized Autocorrelation, also known as the **Autocorrelation Function (ACF)** is defined for a WSS signal as

$$\rho_x(\ell) = \frac{\gamma_x(\ell)}{\gamma_x(0)} = \frac{\gamma_x(\ell)}{\sigma_x^2}$$

where $\gamma_x(\ell)$ is the autocovariance of $x(n)$,

$$\gamma_x(\ell) = \text{E} [[x(n + \ell) - \mu_x][x(n) - \mu_x]^*] = r_x(\ell) - |\mu_x|^2$$

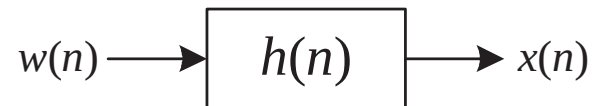
Autocorrelation Function Properties and Examples

$$\rho_x(\ell) = \frac{\gamma_x(\ell)}{\gamma_x(0)} = \frac{\gamma_x(\ell)}{\sigma_x^2}$$

The ACF has a number of useful properties

- Bounded: $-1 \leq \rho_x(\ell) \leq 1$
- White noise, $x(n) \sim \text{WN}(\mu_x, \sigma_x^2)$: $\rho_x(\ell) = \delta(\ell)$
- These enable us to assign meaning to estimated values from signals
- For example,
 - If $\hat{\rho}_x(\ell) \approx \delta(\ell)$, we can conclude that the process consists of nearly uncorrelated samples

Introduction



- Many researchers use signal models to analyze stationary univariate time series
- Goal: estimate the process from which the signal was generated
- Called **signal modeling**
- Related to, but different from, **system identification**
- Popular assumptions:
 - $x(n)$ is ergodic and WSS
 - The system is LTI and stable
 - The input signal is WN
 - The input signal is Gaussian
 - The system is a finite-order PZ system

Terminology

- This text distinguishes between systems and the sequences (processes) that result when a WN input is applied
- Systems: AZ, AP, PZ
- Processes
 - **Moving Average (MA)**
 - **Autoregressive Moving-Average (ARMA)**
 - **Autoregressive (AR)**
- The processes are assumed to have a WN input signal:
 $x(n) \sim \text{WN}(0, \sigma_w^2)$
- In some cases, the input is assumed to be a sum of sinusoids
- In this case, the output consists of line spectra
- Goal: estimate frequencies and magnitudes of spectral components

Example 1: 1st Order Moving Average

Find the autocorrelation function of a 1st order moving average process, MA(1):

$$x(n) = w(n) + b_1 w(n-1)$$

where $w(n) \sim \text{WN}(0, \sigma_w^2)$.

Example 2: 1st Order Autoregressive

Find the autocorrelation function of a 1st order autoregressive process, AR(1):

$$x(n) = -a_1 x(n-1) + w(n)$$

where $w(n) \sim \text{WN}(0, \sigma_w^2)$. Hint: $-\alpha^n u(-n-1) \xleftrightarrow{\mathcal{Z}} \frac{1}{1-\alpha z^{-1}}$ for an ROC of $|z| < |\alpha|$.

Autocorrelation Function Properties

$$\rho_x(\ell) = \frac{\gamma_x(\ell)}{\gamma_x(0)} = \frac{\gamma_x(\ell)}{\sigma_x^2}$$

- In general, the ACF of an $AR(P)$ process decays as a sum of damped exponentials (infinite extent)
- If the $AR(P)$ coefficients are known, the ACF can be determined by solving a set of linear equations
- The ACF of a $MA(Q)$ process is finite: $\rho_x(\ell) = 0$ for $\ell > Q$
- Thus, if the estimated ACF is very small for large lags a $MA(Q)$ model *may* be appropriate
- The ACF of a $ARMA(P, Q)$ process is also a sum of damped exponentials (infinite extent)
- It is difficult to solve for in general

Partial Autocorrelation

Define $P[x(n)|x(1), \dots, x(n-1)]$ as the minimum mean square error linear predictor of $x(n)$ given $\{x(1), \dots, x(n-1)\}$

$$\hat{x}(n) = P[x(n)|x(n-1), \dots, x(1)] = \sum_{k=1}^{n-1} c_k x(n-k)$$

where

$$c_k = \operatorname{argmin}_{c_k} \mathbb{E} \left[(x(n) - \hat{x}(n))^2 \right]$$

Similarly define $P[x(0)|x(1), \dots, x(n-1)]$ as the minimum mean square error linear predictor of $x(0)$ given $\{x(1), \dots, x(n-1)\}$,

$$\hat{x}(0) = P[x(0)|x(n-1), \dots, x(1)] = \sum_{k=1}^{n-1} d_k x(n-k)$$

Partial Autocorrelation: Alternative Definition & Properties

Then the PACF can be defined as the correlation between the residuals

$$\tilde{x}_n(n) \triangleq x(n) - \hat{x}_{1:n-1}(n) = x(n) - P[x(n)|x(n-1), \dots, x(1)]$$

$$\tilde{x}_n(0) \triangleq x(0) - \hat{x}_{1:n-1}(0) = x(0) - P[x(0)|x(n-1), \dots, x(1)]$$

$$\begin{aligned} \alpha(\ell) &\triangleq \frac{E[(x(\ell) - \hat{x}_n(\ell))(x(0) - \hat{x}_n(0))]}{E[(x(0) - \hat{x}_n(0))^2]} \\ &= \frac{E[(x(\ell) - \hat{x}_n(\ell))(x(0) - \hat{x}_n(0))]}{E[(x(n) - \hat{x}_n(n))^2]} \end{aligned}$$

Partial Autocorrelation: Alternative Definition & Properties

Then the PACF can be defined as the correlation between the residuals

$$\alpha(\ell) = \frac{\text{E} [(x(\ell) - \hat{x}_n(\ell)) (x(0) - \hat{x}_n(0))]}{\text{E} [(x(n) - \hat{x}_n(n))^2]}$$

- **Partial Autocorrelation Function (PACF)** also known as, the *partial autocorrelation sequence (PACS)*
- One can think of the PACF as a measure of the correlation of what has not already been explained (the residuals)
- Like the ACF, it depends only on second order properties
- It is a dual of the ACF and has a number of useful and complimentary properties

Partial Autocorrelation Properties & Intuiting

$$\alpha(\ell) \triangleq \begin{cases} 1 & \ell = 0 \\ a_{\ell}^{(\ell)} & \ell > 0 \\ \alpha^*(-\ell) & \ell < 0 \end{cases}$$

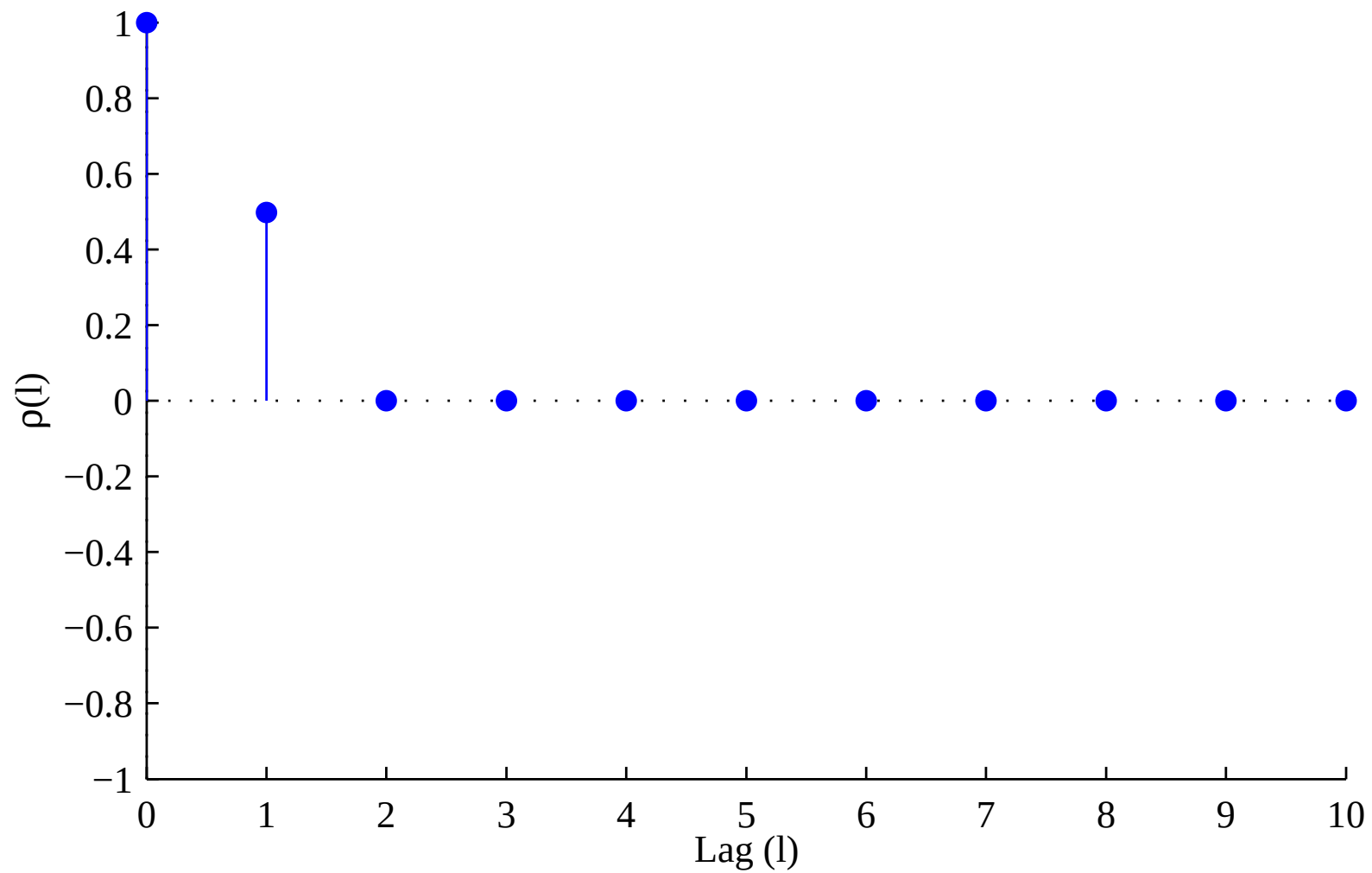
- Intuitively you might expect $|\alpha(\ell)| < |\rho(\ell)|$, but this is not true in general
- Like $\rho(\ell)$, the PACF is bounded: $-1 \leq \alpha(\ell) \leq 1$
- White noise, $x(n) \sim \text{WN}(0, \sigma_x^2)$: $\rho_x(\ell) = \delta(\ell)$
- The PACF of a $\text{AR}(P)$ process is finite: $\alpha_x(\ell) = 0$ for $\ell > P$
- Thus, if the estimated PACF is very small for large lags a $\text{AR}(P)$ model *may* be appropriate
- Surprisingly, the PACF is an infinite sequence for $\text{MA}(Q)$ processes and $\text{ARMA}(P, Q)$ processes

Example 3: MA(1) ACF and PACF

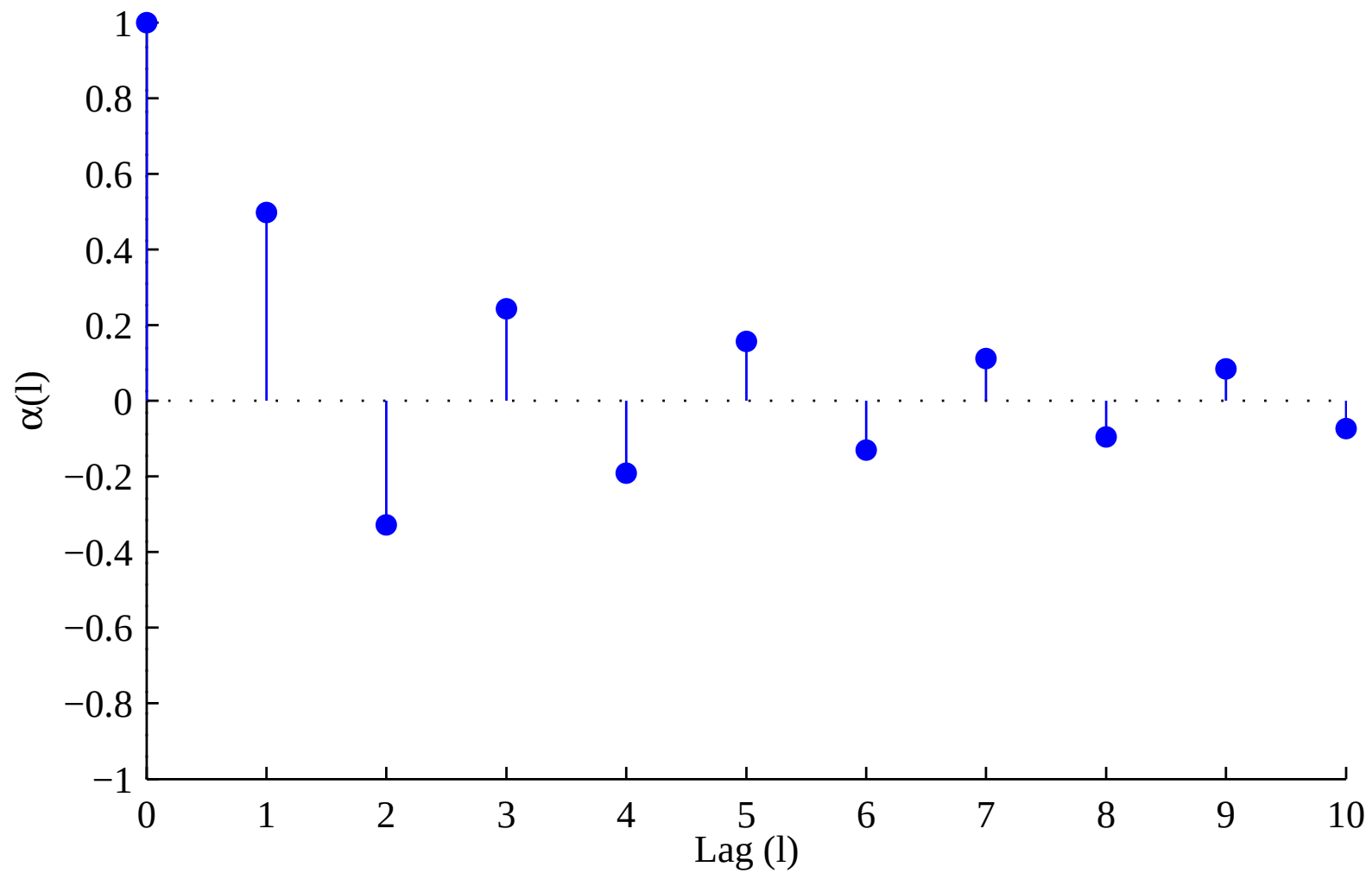
Plot the ACF and PACF of a MA(1) model with $b_1 = 0.9$. Hint: the true PACF is given by

$$\alpha(\ell) = \frac{(-b_1)^\ell (1 - b_1^2)}{1 - b_1^{2(\ell+1)}}$$

Example 3: MA(1) ACF



Example 3: MA(1) PACF



Example 3: MATLAB Code

```
function [] = MovingAverage();
%close all;
%clear all;

L = 10; % Length of autocorrelation calculated
b1 = 0.9; % Coefficient
sw = 1; % White noise power
ac = sw*[(1+b1^2);b1;zeros(L-1,1)]; % Autocovariance = autocorrelation

%=====
% Plot the Autocorrelation
%=====
figure
FigureSet(1,'LTX');
l = 0:L;
acf = ac/ac(1);
h = stem(l,acf);
set(h(1),'MarkerFaceColor','b');
set(h(1),'MarkerSize',4);
ylabel('\rho(l)');
xlabel('Lag (l)');
xlim([0 L]);
ylim([-1 1]);
box off;
AxisLines;
AxisSet(8);
print -depsc MAAutocorrelation;
```

```

%=====
% Calculate the Partial Autocorrelation Function
%=====

pc = zeros(L+1,1);
mc = zeros(L+1,1);
pv = zeros(L+1,1);

pc(1) = 1;
mc(1) = 1;
pv(1) = ac(1);

pc(2) = ac(2)/ac(1);
mc(2) = pc(2);
pv(2) = ac(1)*(1-pc(2).^2);

for c1 = 3:L+1,
    pc(c1) = (ac(c1) - mc(2:c1-1).'*ac((c1-1):-1:2))/pv(c1-1);
    mc(2:c1-1) = mc(2:c1-1) - pc(c1)*mc(c1-1:-1:2);
    mc(c1) = pc(c1);
    pv(c1) = pv(c1-1)*(1-pc(c1).^2);
end;

%=====
% Plot the Partial Autocorrelation
%=====

figure
FigureSet(1,'LTX');
l = 0:L;
h = stem(l,pc);
set(h(1),'MarkerFaceColor','b');

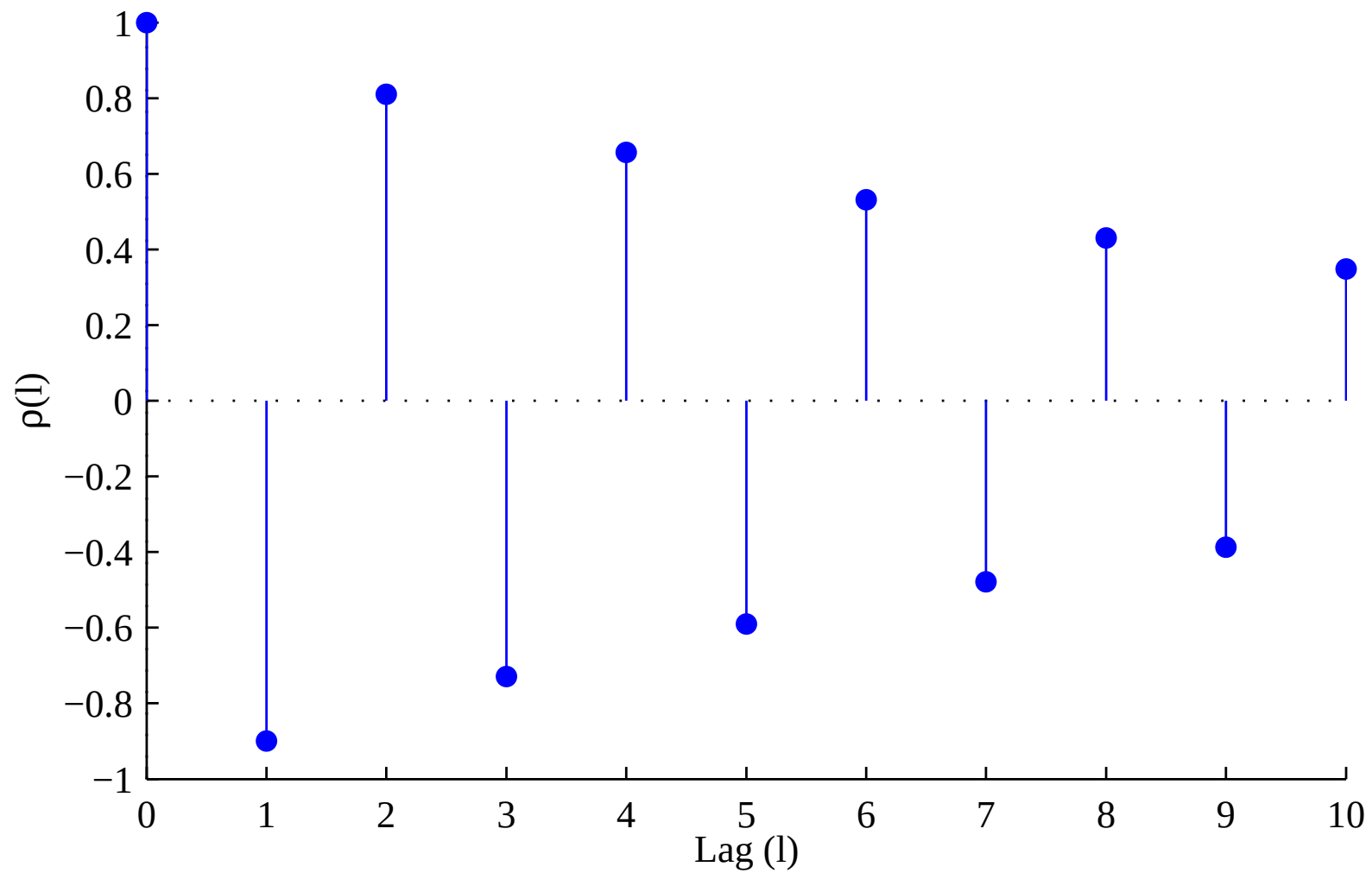
```

```
set(h(1),'MarkerSize',4);  
ylabel('\alpha(l)');  
xlabel('Lag (l)');  
xlim([0 L]);  
ylim([-1 1]);  
box off;  
AxisLines;  
AxisSet(8);  
print -depsc MPartialAutocorrelation;
```

Example 4: AR(1) ACF and PACF

Plot the ACF and PACF of a AR(1) process with $a = [1 \ 0.9]$.

Example 4: AR(1) ACF



Example 4: Relevant MATLAB Code

```
function [] = Autoregressive();
%close all;
%clear all;

L = 10;                                % Length of autocorrelation calculated
a1 = 0.9;                              % Coefficient
sw = 1;                                % White noise power
ac = zeros(L+1,1);

ac(1) = sw/(1-a1^2);
for c1=2:L+1,
    ac(c1) = -a1*ac(c1-1);
end;

%=====
% Plot the Autocorrelation
%=====

figure
FigureSet(1,'LTX');
l = 0:L;
acf = ac/ac(1);
h = stem(l,acf);
set(h(1),'MarkerFaceColor','b');
set(h(1),'MarkerSize',4);
ylabel('\rho(l)');
xlabel('Lag (l)');
xlim([0 L]);
```

```

ylim([-1 1]);
box off;
AxisLines;
AxisSet(8);
print -depsc ARAutocorrelation;

%=====
% Calculate the Partial Autocorrelation Function
%=====
pc = zeros(L+1,1);
mc = zeros(L+1,1);
pv = zeros(L+1,1);

pc(1) = 1;
mc(1) = 1;
pv(1) = ac(1);

pc(2) = ac(2)/ac(1);
mc(2) = pc(2);
pv(2) = ac(1)*(1-pc(2).^2);

for c1 = 3:L+1,
    pc(c1) = (ac(c1) - mc(2:c1-1).'*ac((c1-1):-1:2))/pv(c1-1);
    mc(2:c1-1) = mc(2:c1-1) - pc(c1)*mc(c1-1:-1:2);
    mc(c1) = pc(c1);
    pv(c1) = pv(c1-1)*(1-pc(c1).^2);
end;

%=====
% Plot the Partial Autocorrelation
%=====

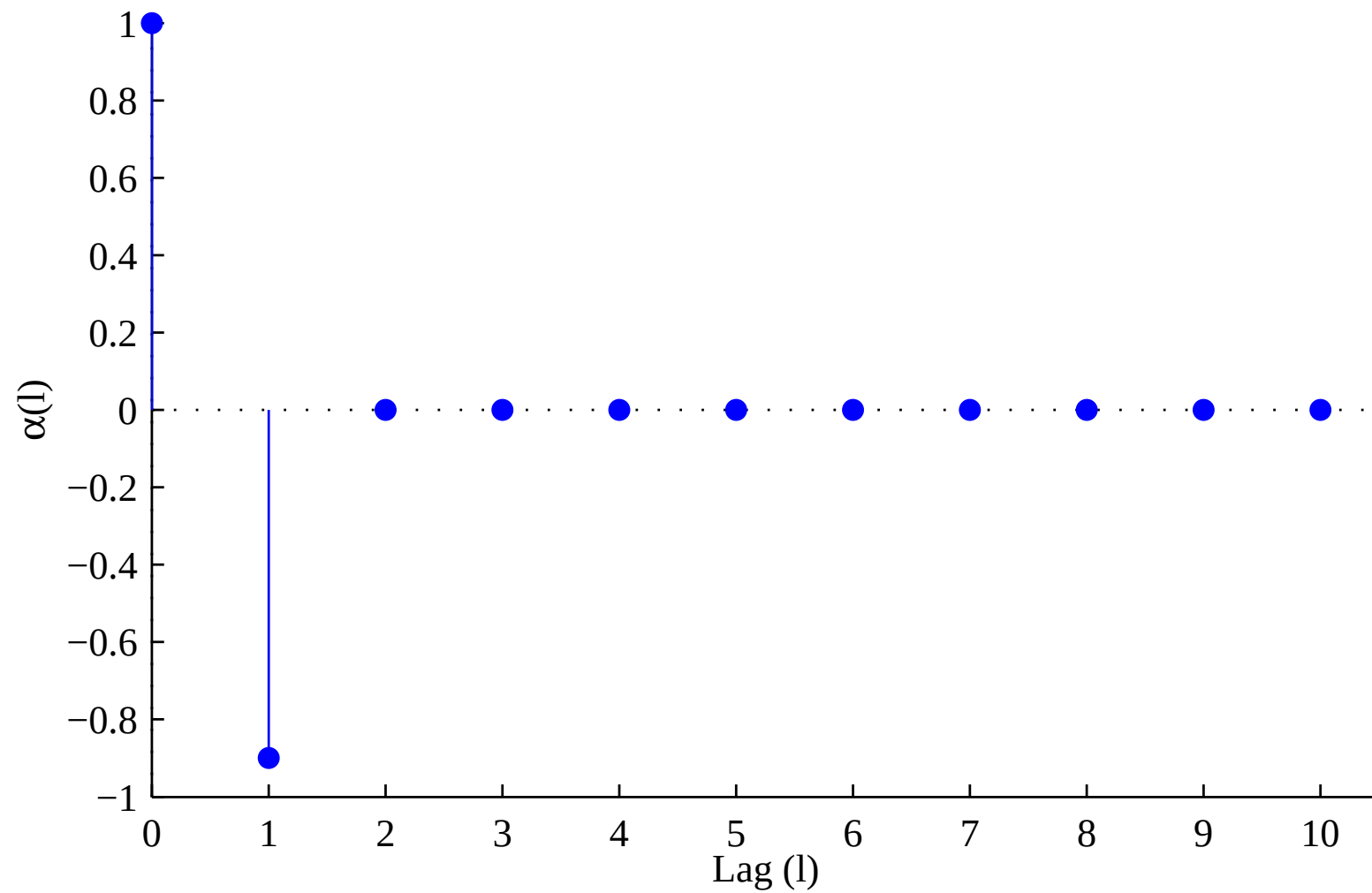
```

```

figure
FigureSet(1,'LTX');
l = 0:L;
h = stem(l,pc);
set(h(1),'MarkerFaceColor','b');
set(h(1),'MarkerSize',4);
ylabel('\alpha(l)');
xlabel('Lag (l)');
xlim([0 L+0.5]);
ylim([-1 1]);
box off;
AxisLines;
AxisSet(8);
print -depsc ARPartialAutocorrelation;

```

Example 4: AR(1) PACF



Autocovariance Estimation

- We've seen that the second-order statistics are a handy, though incomplete, characterization of WSS stochastic processes
- We would like to *estimate* these properties from realizations
 - Single signal: $\gamma_x(\ell)$, $r_x(\ell)$, $\alpha_x(\ell)$, $R_x(e^{j\omega})$
 - Two or more signals: $\gamma_{yx}(\ell)$, $r_{yx}(\ell)$, $R_{yx}(\ell)$, $\mathcal{G}_{yx}^2(e^{j\omega})$
- What are the best estimators?

Autocovariance Estimation Options

In practical applications, we only have a real finite data record $\{x(n)\}_0^{N-1}$. There are two popular estimators of autocovariance worth considering: “unbiased” and biased.

“Unbiased”

$$\hat{\gamma}_u(\ell) \triangleq \frac{1}{N - |\ell|} \sum_{n=0}^{N-1-|\ell|} [x(n + |\ell|) - \hat{\mu}_x] [x(n) - \hat{\mu}_x] \quad |\ell| < N$$

and $\hat{\gamma}_u(\ell) = 0$ for $|\ell| \geq N$. Here $\hat{\mu}_x$ is the sample average of the sequence defined as

$$\hat{\mu}_x \triangleq \frac{1}{N} \sum_{n=0}^{N-1} x(n)$$

Unbiased Autocovariance Estimation

$$\hat{\gamma}_u(\ell) \triangleq \frac{1}{N - |\ell|} \sum_{n=0}^{N-1-|\ell|} [x(n + |\ell|) - \hat{\mu}_x] [x(n) - \hat{\mu}_x]$$

- Discussed briefly in the book
- The estimate has even symmetry: $\hat{\gamma}_u(\ell) = \hat{\gamma}_u(-\ell)$
- At longer lags, we have fewer terms to estimate the autocovariance
- We have no way to estimate $\gamma(\ell)$ for $|\ell| \geq N$
- We know that each pair $\{x(n + |\ell|), x(n)\}$ for all n have the same distribution because the process is assumed WSS and ergodic
- This is a natural estimator that we know converges asymptotically ($N \rightarrow \infty$)

Unbiased Autocovariance Estimation

$$\hat{\gamma}_u(\ell) \triangleq \frac{1}{N - |\ell|} \sum_{n=0}^{N-1-|\ell|} [x(n + |\ell|) - \hat{\mu}_x] [x(n) - \hat{\mu}_x]$$

- If we used the true mean μ_x instead of $\hat{\mu}_x$, $\hat{\gamma}_u(\ell)$ would be *unbiased*
- When we use $\hat{\mu}_x$ the estimate is asymptotically unbiased
- The bias is $\mathcal{O}(1/N)$
- Much smaller than the variance, so it may be ignored

Biased Autocovariance Estimation

$$\begin{aligned}\hat{\gamma}_b(\ell) &\triangleq \frac{1}{N} \sum_{n=0}^{N-1-|\ell|} [x(n+|\ell|) - \hat{\mu}_x] [x(n) - \hat{\mu}_x] \quad |\ell| < N \\ &= \frac{N-|\ell|}{N} \hat{\gamma}_u(\ell)\end{aligned}$$

- Our book (and most other books) lists a different estimate
- This estimate uses a divisor of N rather than $(N - |\ell|)$
- If we ignore the effect of estimating μ_x , this bias is obvious

$$\mathbb{E}[\hat{\gamma}(\ell)] = \frac{N-|\ell|}{N} \gamma(\ell)$$

- The bias of this estimator is larger than the “unbiased” estimator
- Some claim that in general, the “biased” estimator has a smaller MSE
- The variance, must therefore be much smaller

Biased versus Unbiased Estimators

$$\hat{\gamma}_b(\ell) = \frac{N - |\ell|}{N} \hat{\gamma}_u(\ell) \propto \sum_{n=0}^{N-1-|\ell|} [x(n + |\ell|) - \hat{\mu}_x] [x(n) - \hat{\mu}_x]$$

- The estimators are often called the **sample autocovariance functions**
- Most software and books prefer the biased estimate
- Why prefer a biased estimate to an unbiased estimate?
- Our goal is to estimate the sequence, not just $\gamma(\ell)$ for a specific lag ℓ

Biased versus Unbiased Estimators

$$\hat{\gamma}_b(\ell) = \frac{N - |\ell|}{N} \hat{\gamma}_u(\ell) \propto \sum_{n=0}^{N-1-|\ell|} [x(n + |\ell|) - \hat{\mu}_x] [x(n) - \hat{\mu}_x]$$

- The key advantage of $\hat{\gamma}_b(\ell)$ is that it is *positive semi-definite* (i.e., nonnegative definite)
- There are many reasons why this property is important
 - We know the true autocovariance has this property
 - Autoregressive models built with the positive-definite estimates of $\gamma(\ell)$ are stable
 - Most estimators of power spectral density $R(e^{j\omega})$ are nonnegative if they are based on a positive-definite estimate of $\gamma(\ell)$
- $\hat{\gamma}_u(\ell)$ *may* be positive definite for a particular sequence, but it is not guaranteed in general

Biased versus Unbiased Estimators

$$\hat{\gamma}_b(\ell) = \frac{N - |\ell|}{N} \hat{\gamma}_u(\ell) \propto \sum_{n=0}^{N-1-|\ell|} [x(n + |\ell|) - \hat{\mu}_x] [x(n) - \hat{\mu}_x]$$

- Although $\hat{\gamma}_b(\ell)$ is biased,
 - The bias is small at small lags
 - For large lags, the bias is towards 0: $\hat{\gamma}(\ell) \rightarrow 0$ as $\ell \rightarrow \infty$
 - This is also a property of the true autocorrelation
- If $\gamma(\ell)$ is small for large lags, then the bias is also small
- The biased estimator has considerably less variance at large lags (the tail)

Biased	$\text{var}\{\hat{\gamma}_b(\ell)\} = \mathcal{O}(1/N)$
Unbiased	$\text{var}\{\hat{\gamma}_u(\ell)\} = \mathcal{O}(1/(N - \ell))$

Biased is Better?

$$\hat{\gamma}_b(\ell) = \frac{N - |\ell|}{N} \hat{\gamma}_u(\ell) \propto \sum_{n=0}^{N-1-|\ell|} [x(n + |\ell|) - \hat{\mu}_x] [x(n) - \hat{\mu}_x]$$

- In general
 - At small lags, there is little difference between the two estimators
 - At large lags, the larger bias of the biased model is favorably traded for reduced variance
- In most cases, the biased model has smaller MSE, though it has not been proven rigorously
- For the remainder of the class will use the biased estimator, unless otherwise noted $\hat{\gamma}(\ell) = \hat{\gamma}_b(\ell)$

Estimated Autocorrelation Covariance

$$\hat{\gamma}_b(\ell) = \frac{N - |\ell|}{N} \hat{\gamma}_u(\ell) \propto \sum_{n=0}^{N-1-|\ell|} [x(n + |\ell|) - \hat{\mu}_x] [x(n) - \hat{\mu}_x]$$

- As with all estimators, we would like to have confidence intervals
- These are hard to obtain, in general
- Need more assumptions
 - Stationary up to order four

$$E[x(n)x(n+k)x(n+\ell)x(n+m)] = f(k, \ell, m)$$

- Mean is zero $\mu_x = 0$, so does not need to be estimated

Estimated Autocorrelation Variance

$$\hat{\gamma}_b(\ell) = \hat{r}_b(\ell) = \frac{1}{N} \sum_{n=0}^{N-1-|\ell|} x(n+|\ell|)x(n)$$

$$\hat{\gamma}_u(\ell) = \hat{r}_u(\ell) = \frac{1}{N-|\ell|} \sum_{n=0}^{N-1-|\ell|} x(n+|\ell|)x(n)$$

- The bias is

$$\mathbb{E}[\hat{r}_b(\ell)] = \frac{N-|\ell|}{N} r(\ell) \qquad \mathbb{E}[\hat{r}_u(\ell)] = r(\ell)$$

- The covariance of $\hat{r}(\ell)$ is complicated and not usable in practice
 - Depends on fourth joint cumulant of $\{x(n), x(n+k), x(n+\ell), x(n+m)\}$
 - Depends on true unknown autocorrelation
- If process is Gaussian, then the fourth joint cumulant is zero

Estimated Autocorrelation Variance Continued

If Gaussian random process,

$$\text{var}\{\hat{r}_b(\ell)\} = \frac{1}{N} \sum_{m=-(N+\ell)+1}^{N-\ell-1} \left(\frac{N-|m|+\ell}{N} \right) (r^2(m) + r(m+\ell)r(m-\ell))$$

- The same applies to the unbiased estimate with a divisor of $1/(N - |\ell|)$ instead of $1/N$
- This is still problematic because we don't know the true $r(\ell)$ in most applications
- If we did, we wouldn't need to estimate it!
- This is often what prevents us from making desired inferences about our estimators:
 - Desired properties of the sampling distribution depend on unknown properties of the random process

Estimated ACF

The natural estimate of the ACF is

$$\hat{\rho}_b(\ell) \triangleq \frac{\hat{\gamma}_b(\ell)}{\hat{\gamma}(0)} \qquad \hat{\rho}_u(\ell) \triangleq \frac{\hat{\gamma}_u(\ell)}{\hat{\gamma}(0)}$$

- Same tradeoffs exist between the biased and unbiased estimates
- Also
 - $|\hat{\rho}_b(\ell)| \leq 1$ for all ℓ
 - Not true in general for $\hat{\rho}_u(\ell)$
- They are the same at $\ell = 0$
- Often called the **sample autocorrelation function**
- Again, the bias, covariance, and variance of the estimators is complicated and based on unknown properties

Estimated ACF Variance

Again, if $x(n)$ is a Gaussian process then

$$\begin{aligned} \text{var}\{\hat{\rho}_b(\ell)\} \approx \frac{1}{N} \sum_{m=-\infty}^{\infty} \rho^2(m) + \rho(m+\ell)\rho(m-\ell) \\ + 2\rho^2(\ell)\rho^2(m) - 4\rho(\ell)\rho(m)\rho(m-\ell) \end{aligned}$$

- The fourth cumulant is also absent if $x(n)$ is generated by a linear process with *independent* inputs
- The sample ACF, $\hat{\rho}(\ell)$ will generally have more correlation than the true $\rho(\ell)$
- It will generally be less damped and decay more slowly than $\rho(\ell)$
- Applies to the estimated autocovariance and autocorrelations as well

Confidence Intervals

Let $x(n)$ be an IID sequence. Then

$$\rho(0) = 1$$

$$\rho(\ell) = 0 \quad |\ell| > 0$$

$$\text{cov}\{\hat{\rho}(\ell), \hat{\rho}(\ell + m)\} \approx 0 \quad m \neq 0$$

$$\text{var}\{\hat{\rho}_b(\ell)\} \approx \frac{1}{N} \quad |\ell| > 0$$

$$\text{var}\{\hat{\rho}_u(\ell)\} \approx \frac{N}{(N - |\ell|)^2} \quad |\ell| > 0$$

- In general, it is not possible to obtain confidence intervals for the estimated ACF because the variance of the estimator depends on the true ACF
- Instead, it is common practice to plot the confidence intervals of a purely random process

Confidence Intervals

- If N is large enough, the central limit theorem applies and $\hat{\rho}_b(\ell)$ is approximately normal
- In this case, we can use the Normal cdf to plot confidence intervals of an IID sequence
- These are proportional to $\pm \sqrt{\text{var}\{\hat{\rho}_b(\ell)\}}$

Partial Autocorrelation Estimation

- There are similar issues surrounding partial autocorrelation
- However, in this case we always use the biased estimate of autocorrelation to estimate the PACF
- This is necessary, in this case, to ensure that the AR models are bounded
- Less is known about the statistics of the PACF (mean, variance, and confidence intervals)
- However, for reasons similar to that of the ACF, for a WN process the CLT applies and we can use the same confidence intervals as for the ACF

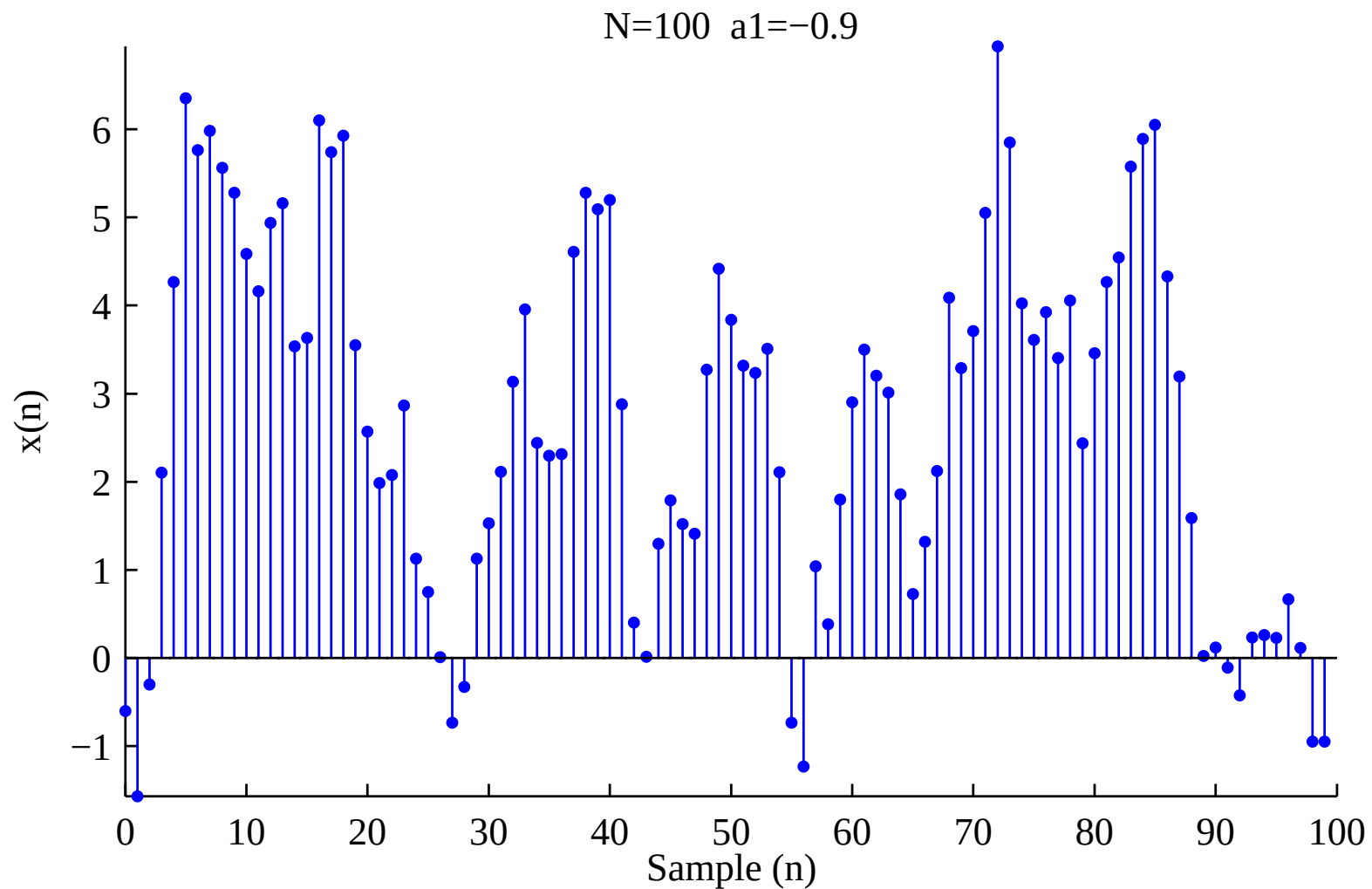
Example 5: 1st Order Autoregressive

Find the autocorrelation function of a 1st order autoregressive process, AR(1):

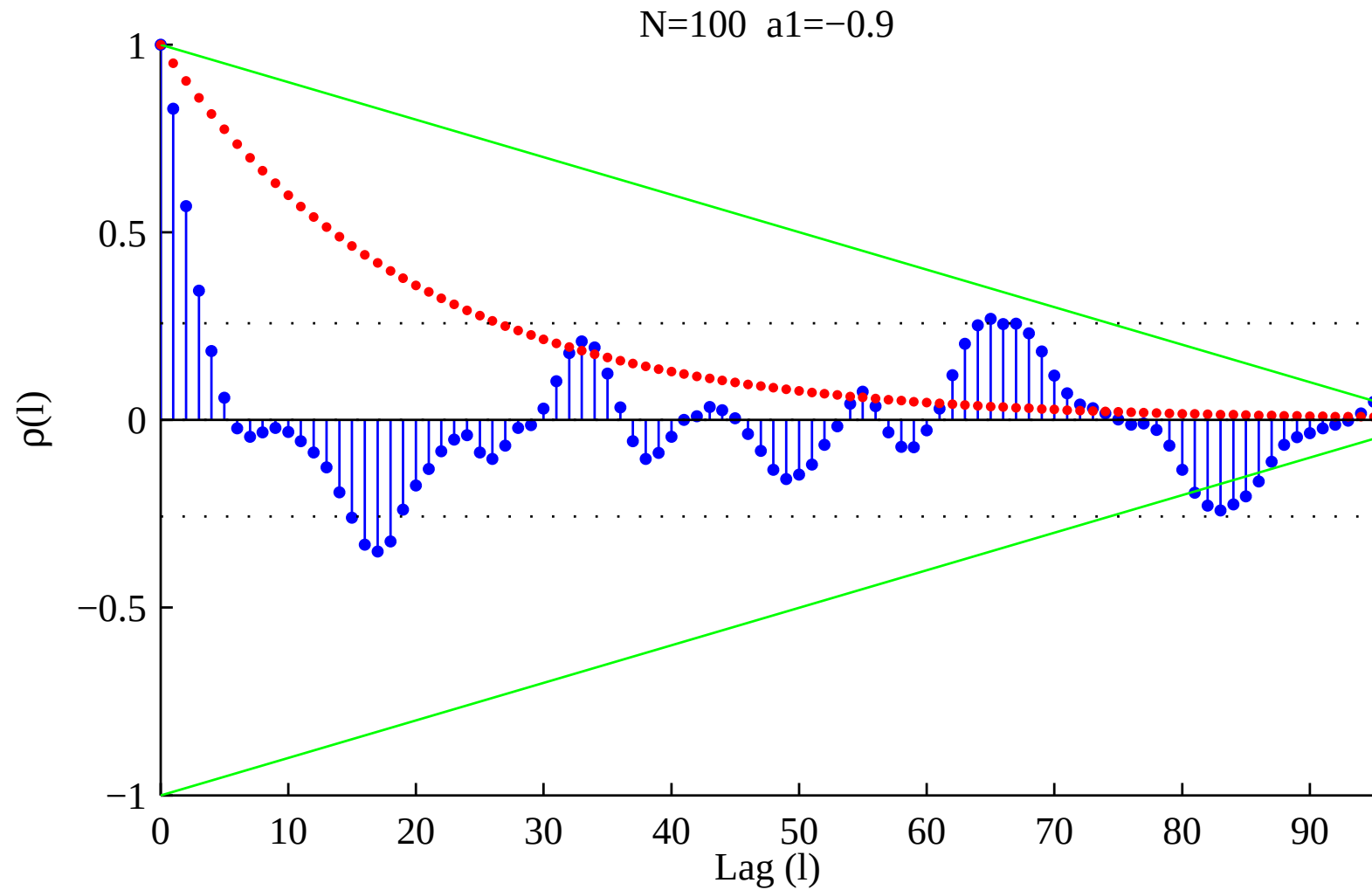
$$x(n) = -a_1 x(n-1) + w(n)$$

where $w(n) \sim \text{WN}(0, \sigma_w^2)$. Estimate the ACF using the biased and unbiased estimates for $N = 100$. Do so several times for different values of a_1 .

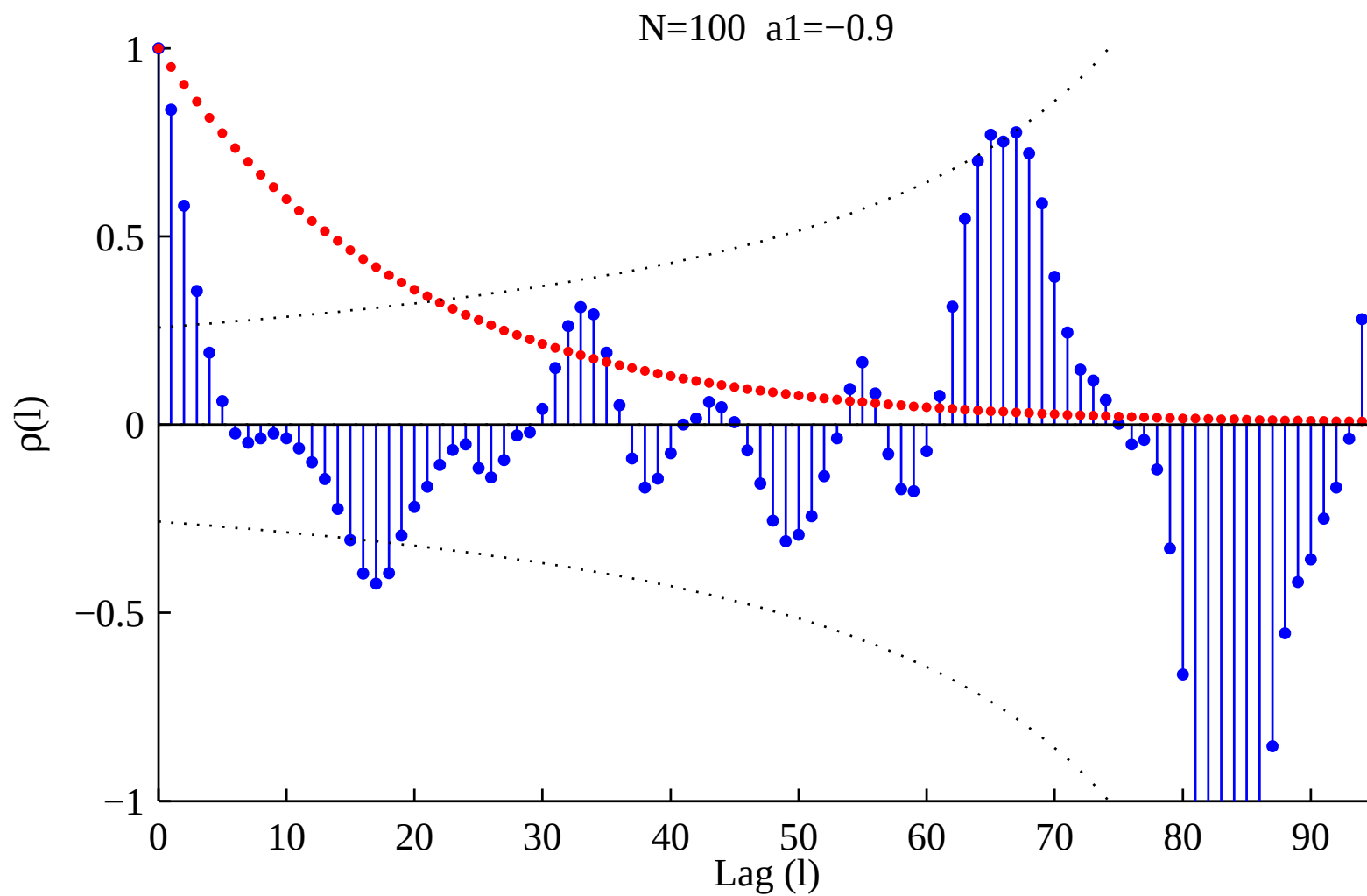
Example 5: AR(1) Signal



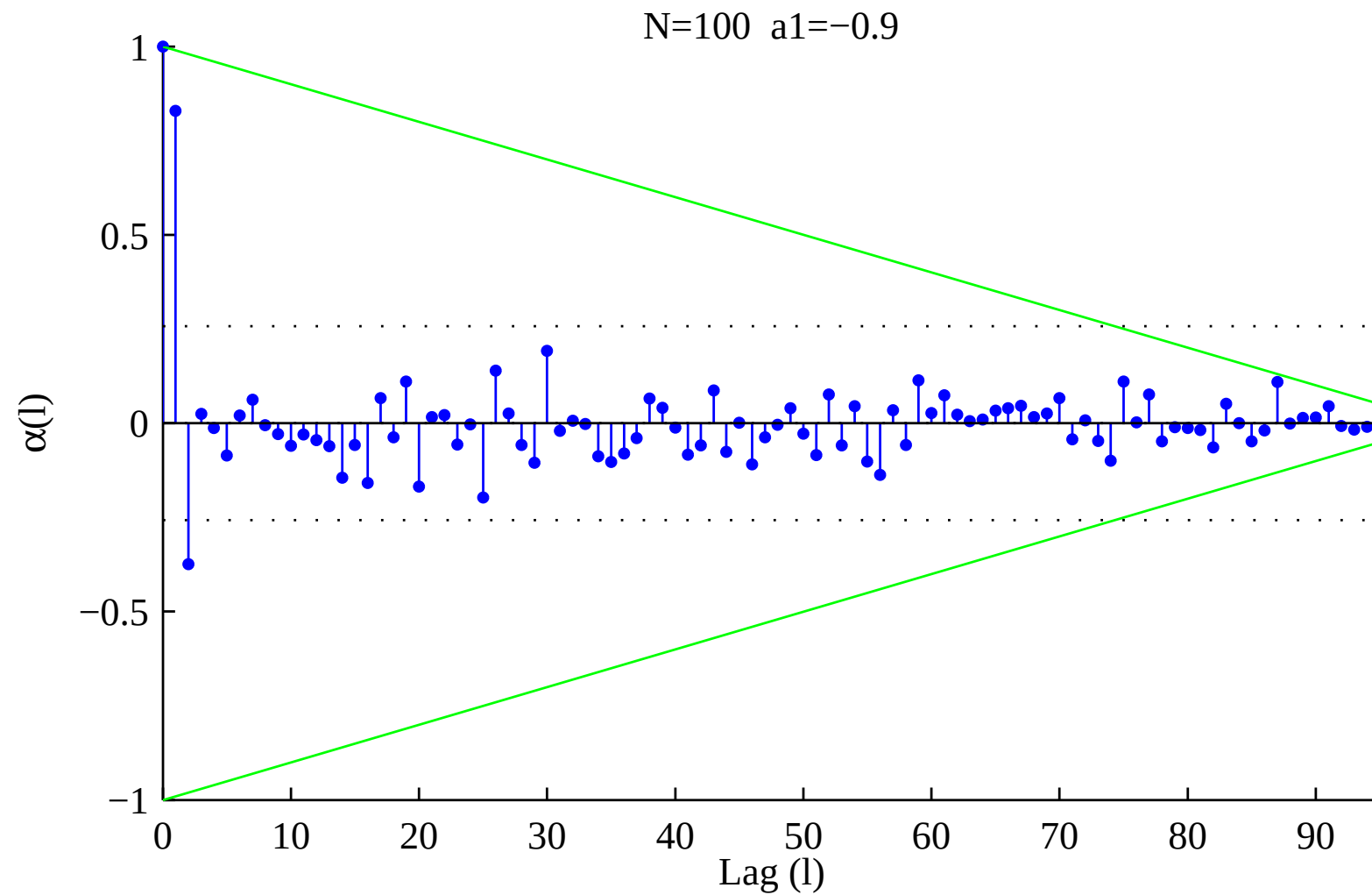
Example 5: AR(1) ACF



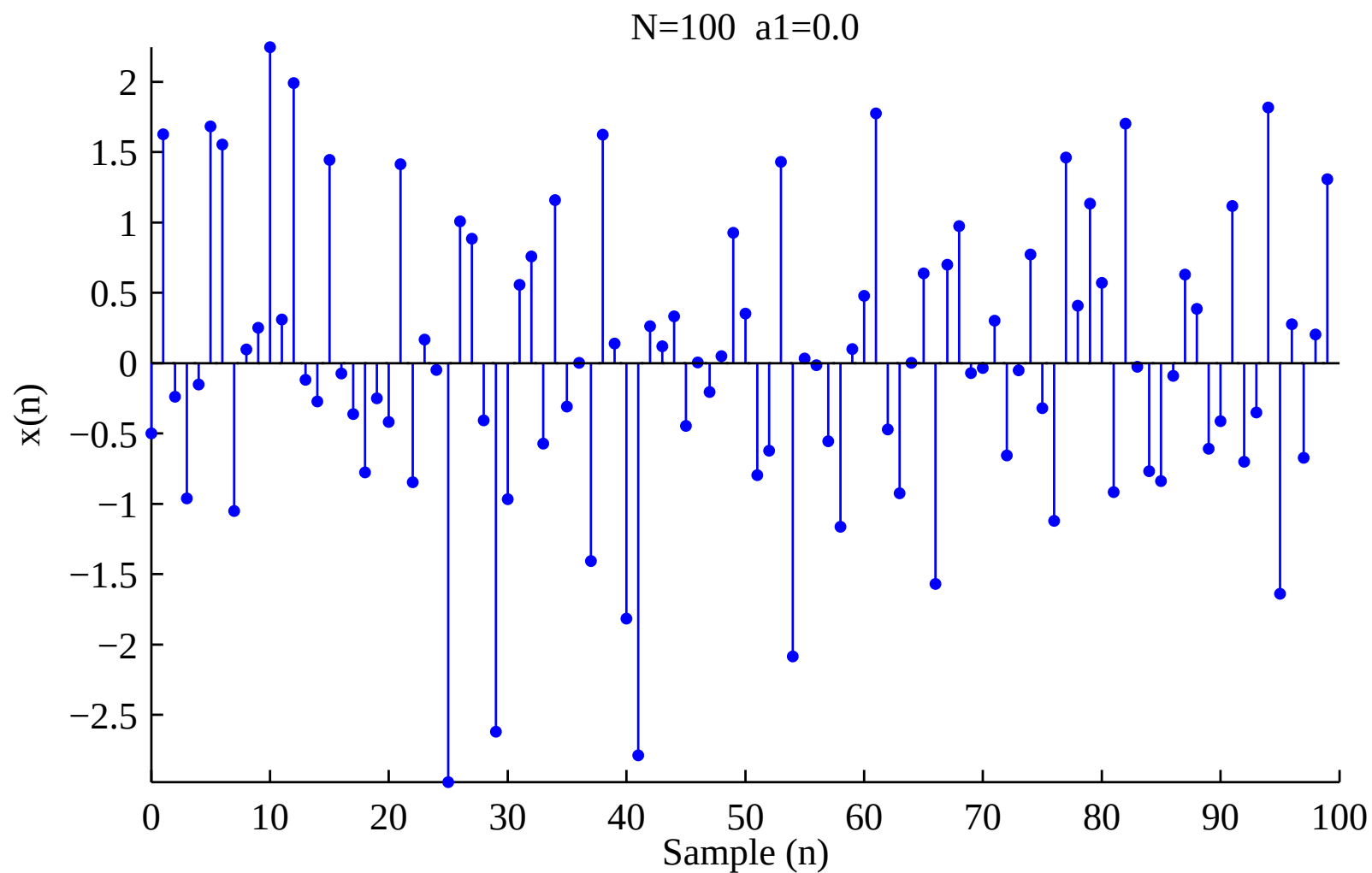
Example 5: AR(1) ACF



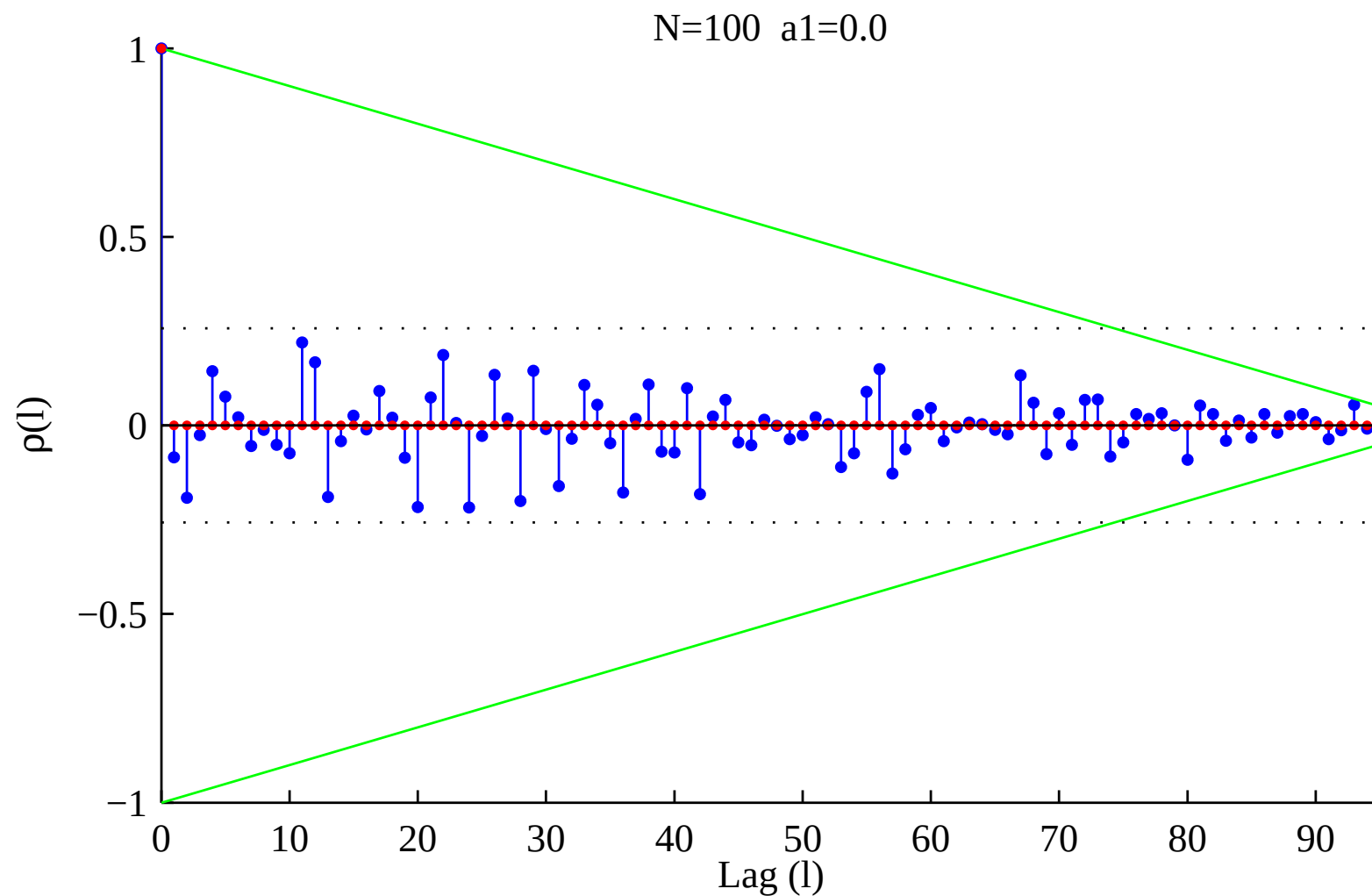
Example 5: AR(1) PACF



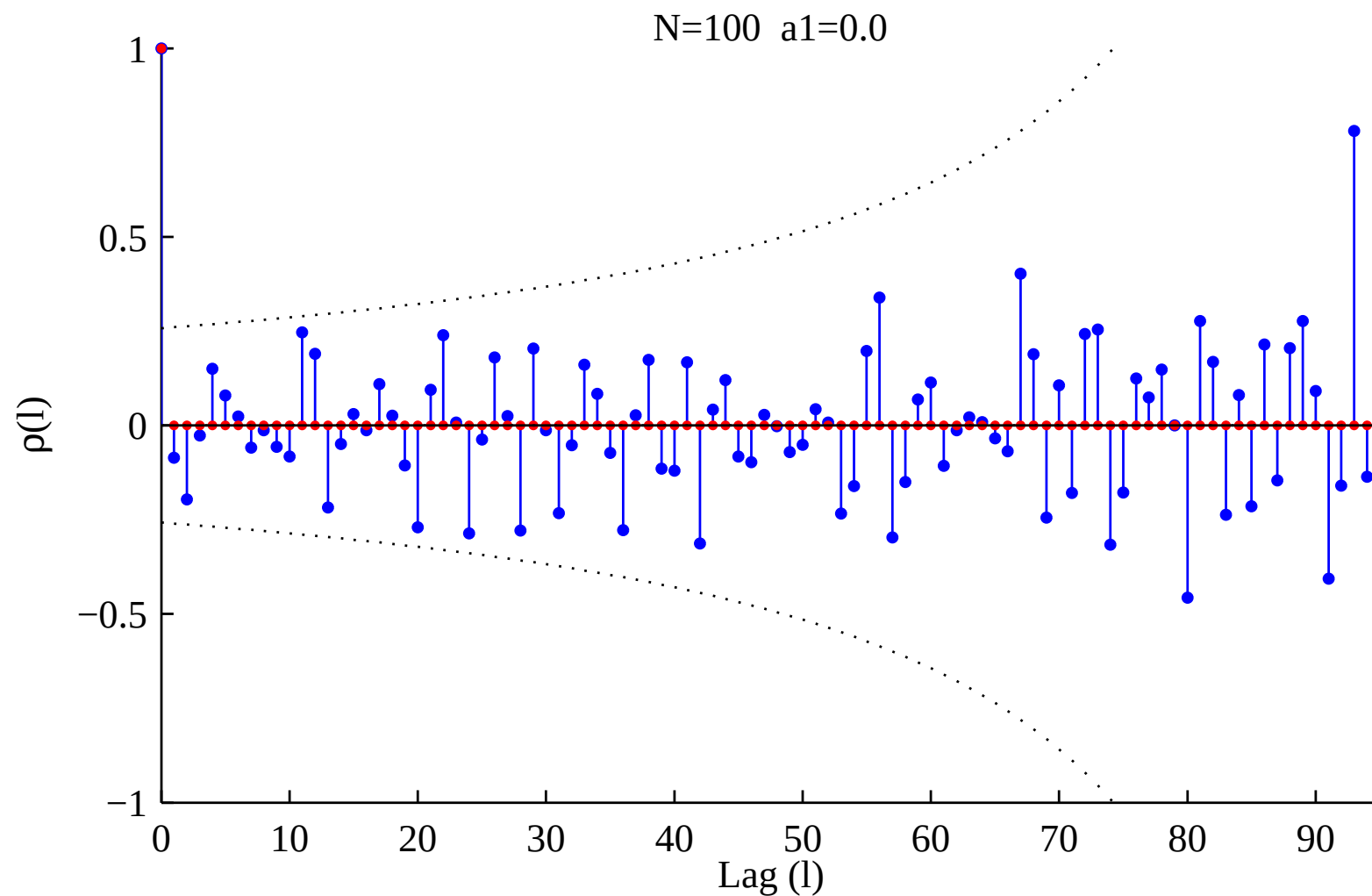
Example 5: AR(1) Signal



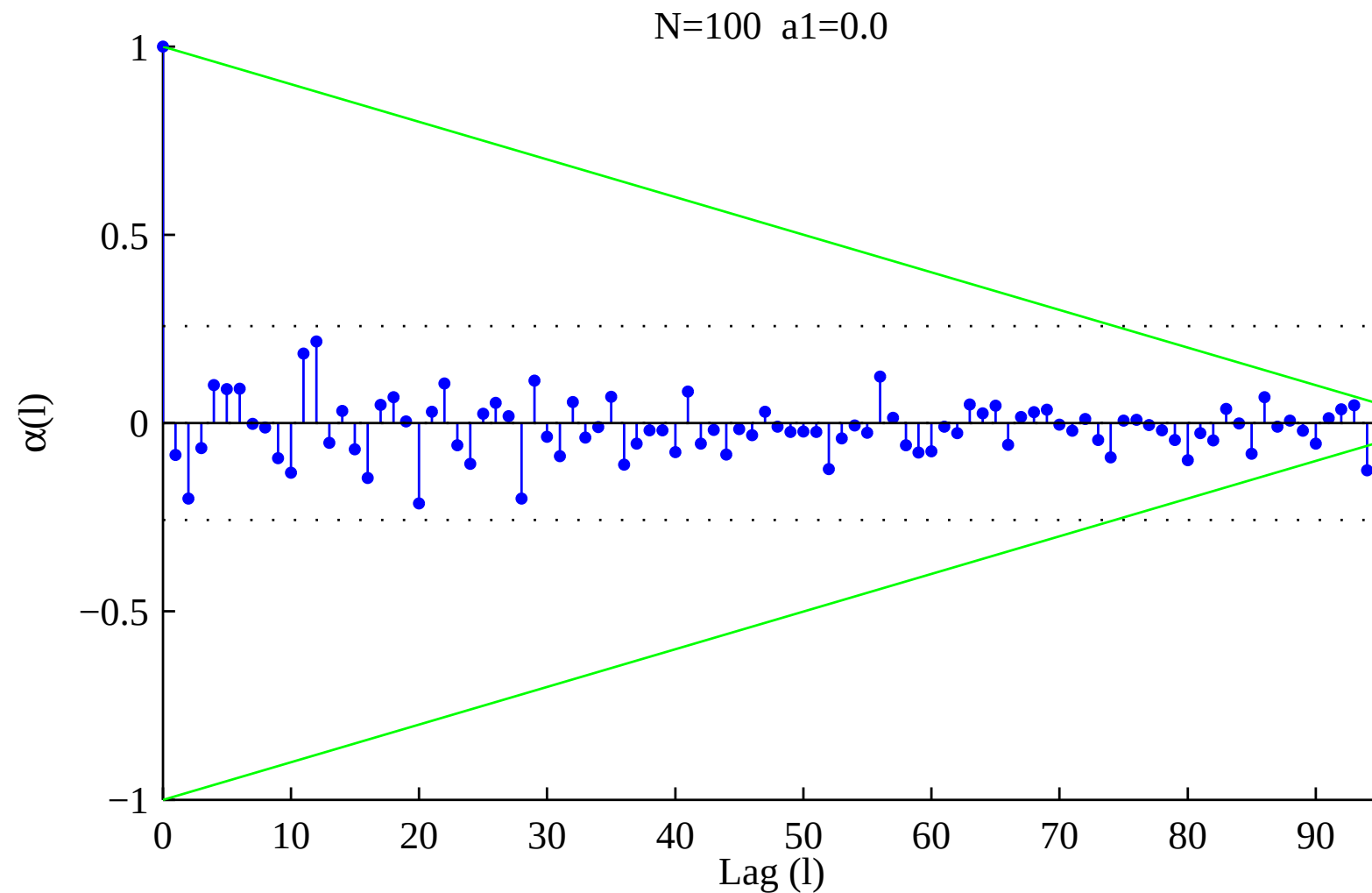
Example 5: AR(1) ACF



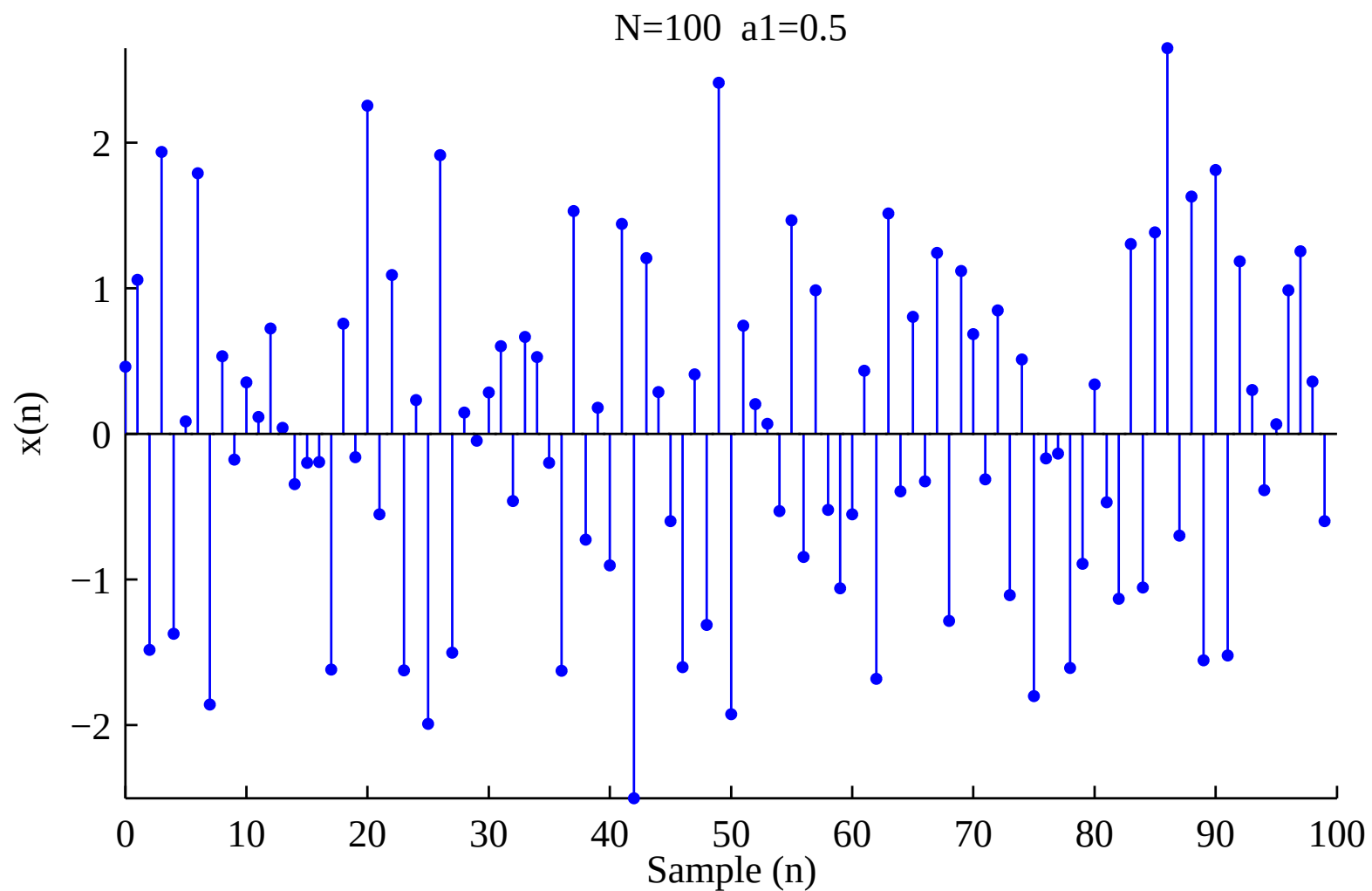
Example 5: AR(1) ACF



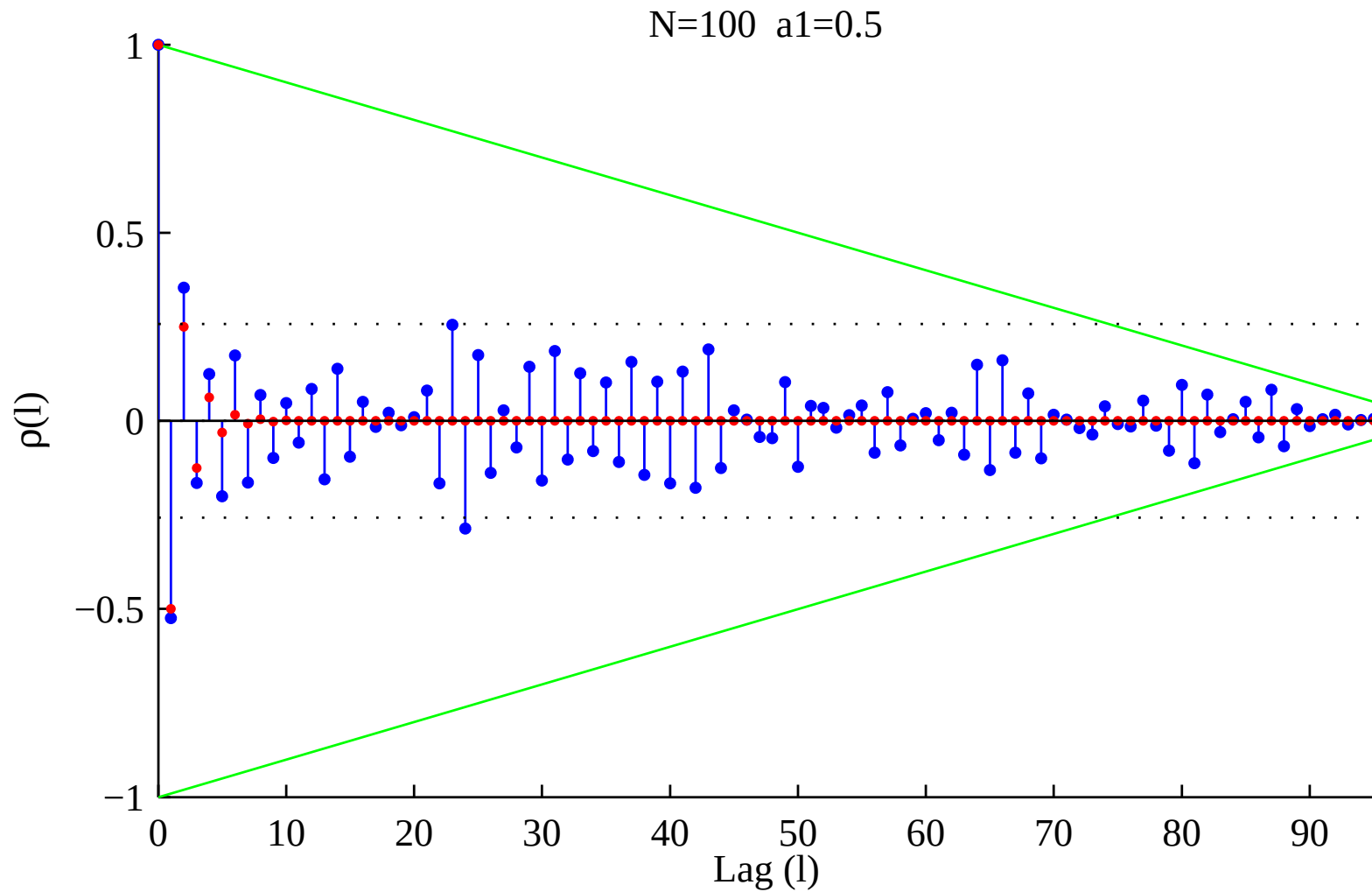
Example 5: AR(1) PACF



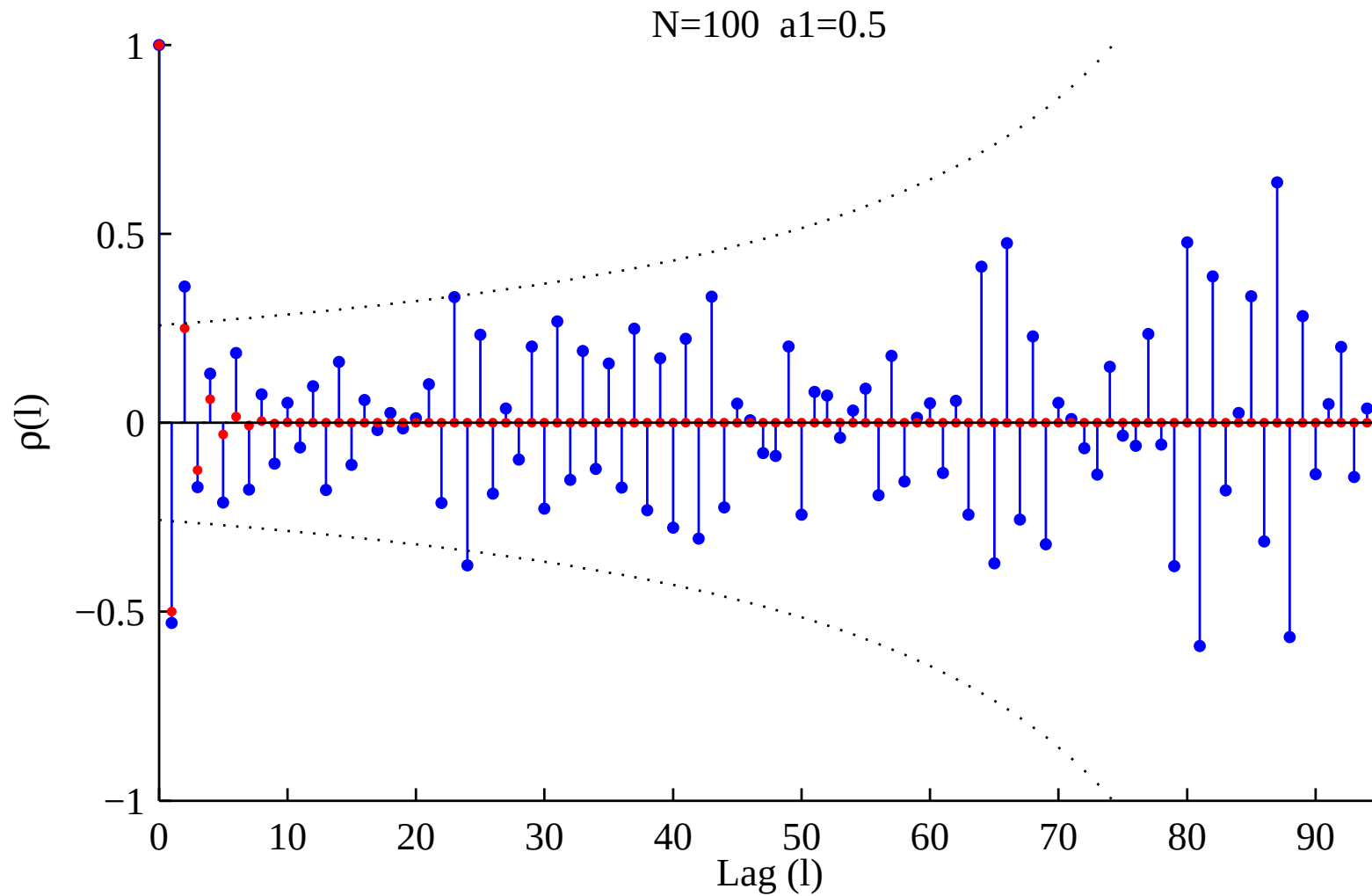
Example 5: AR(1) Signal



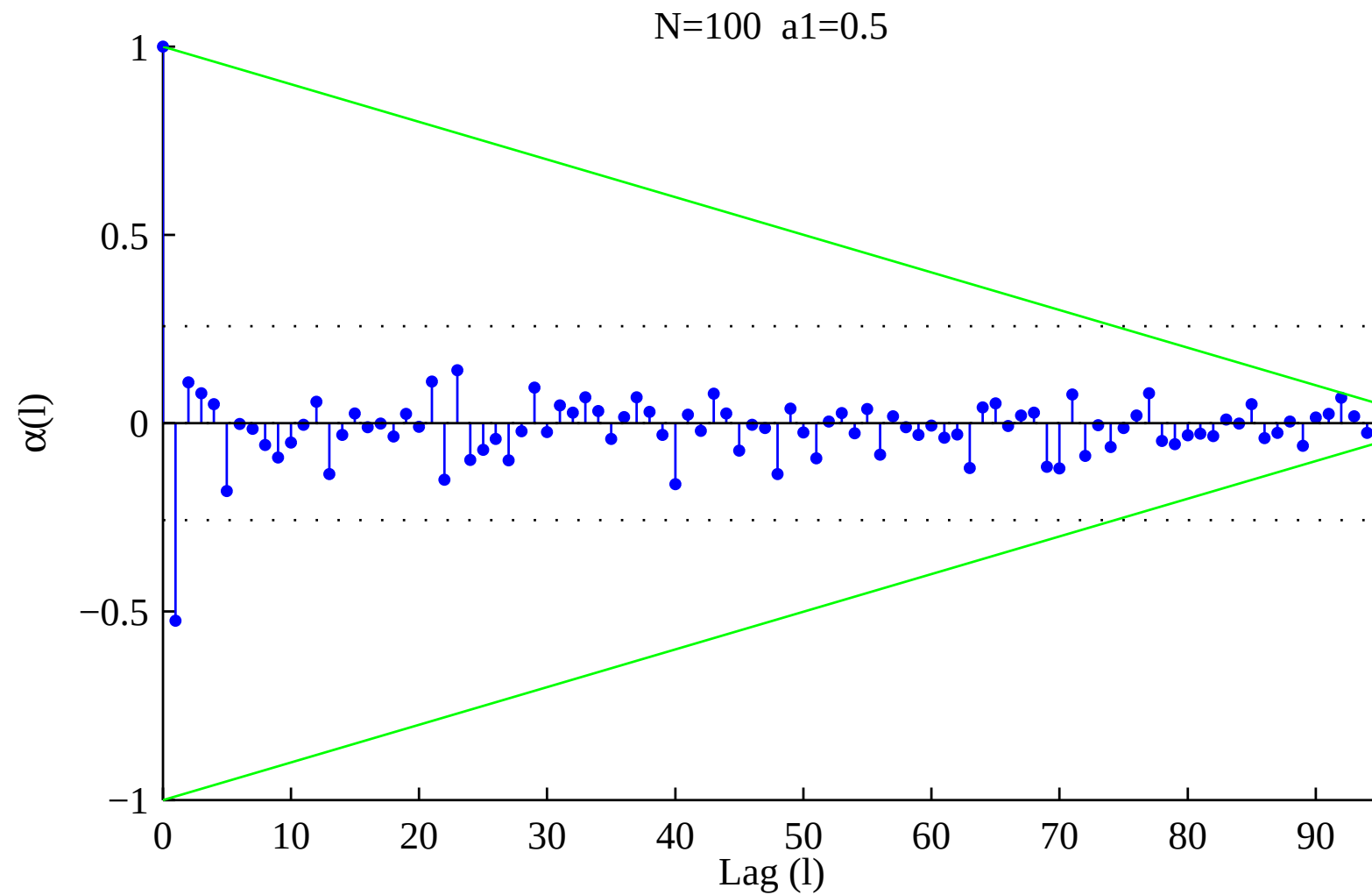
Example 5: AR(1) ACF



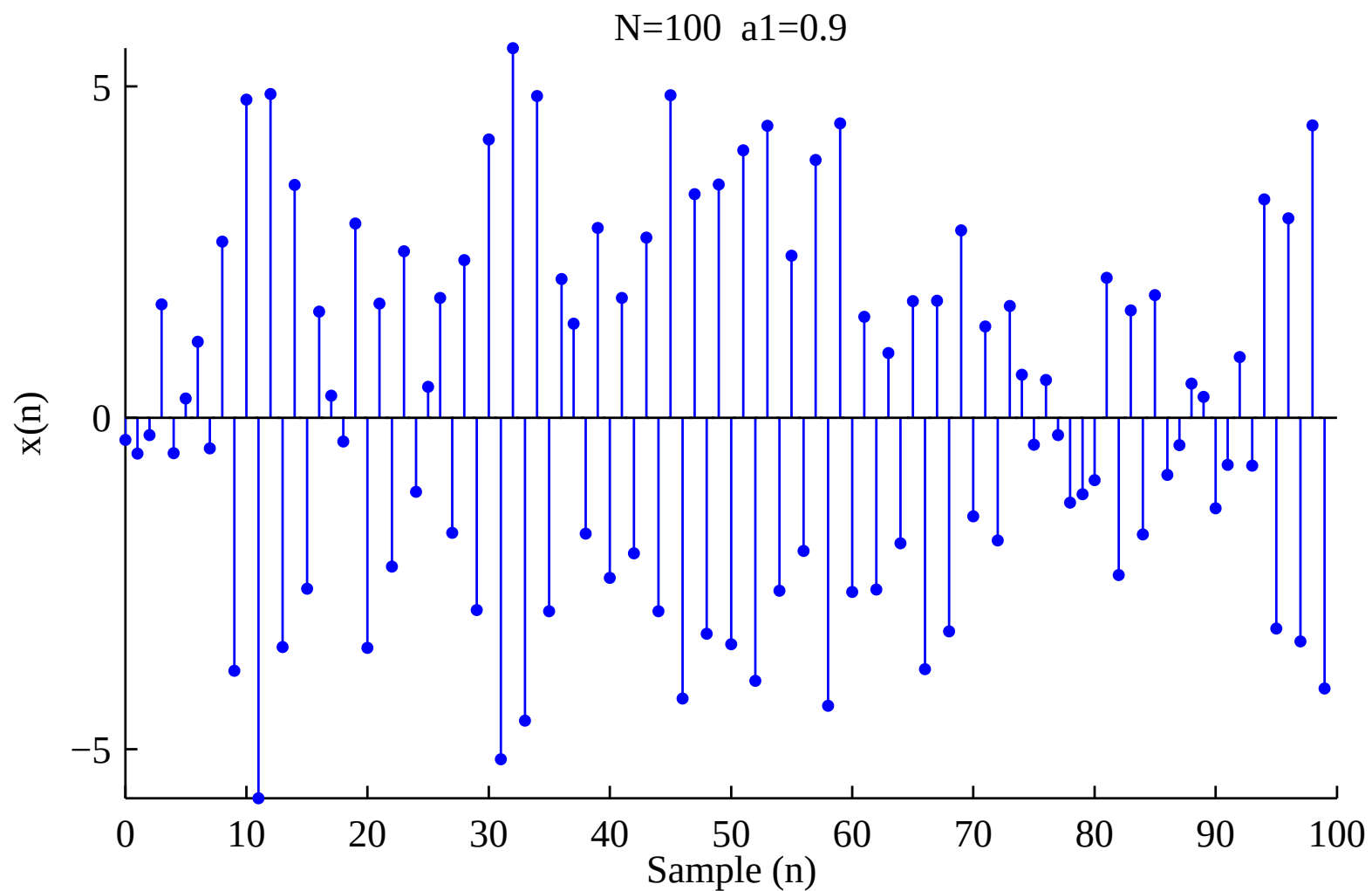
Example 5: AR(1) ACF



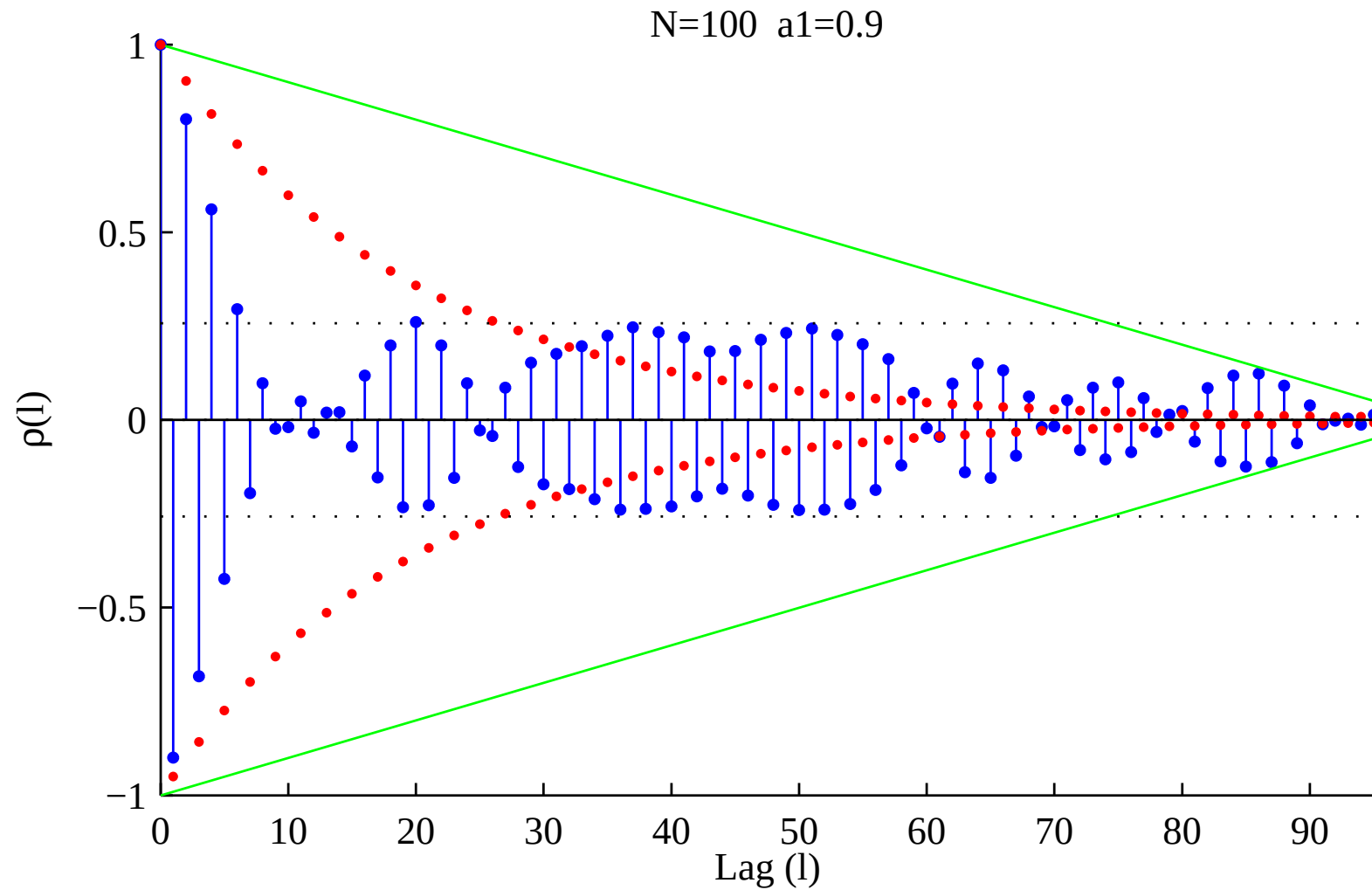
Example 5: AR(1) PACF



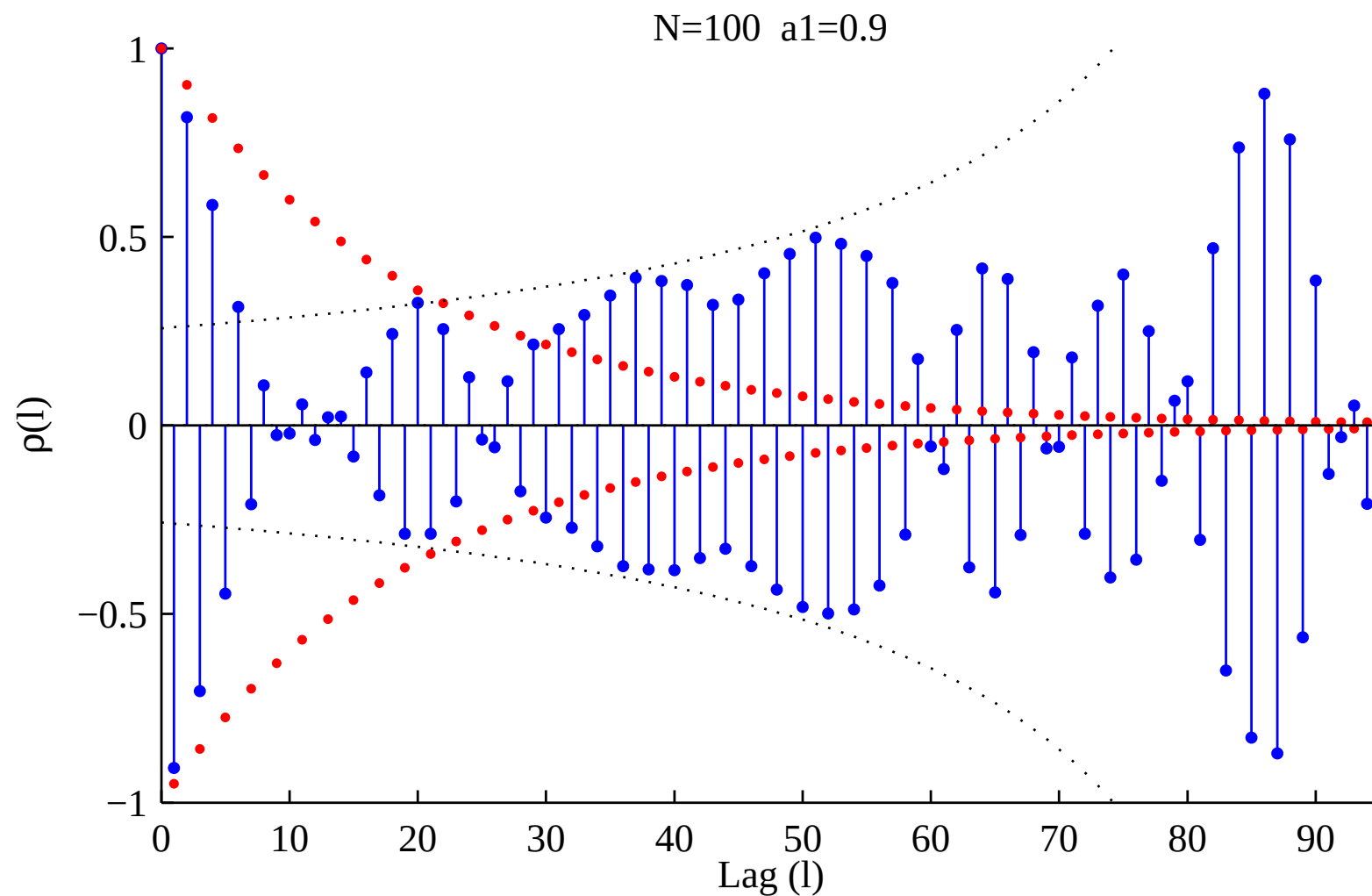
Example 5: AR(1) Signal



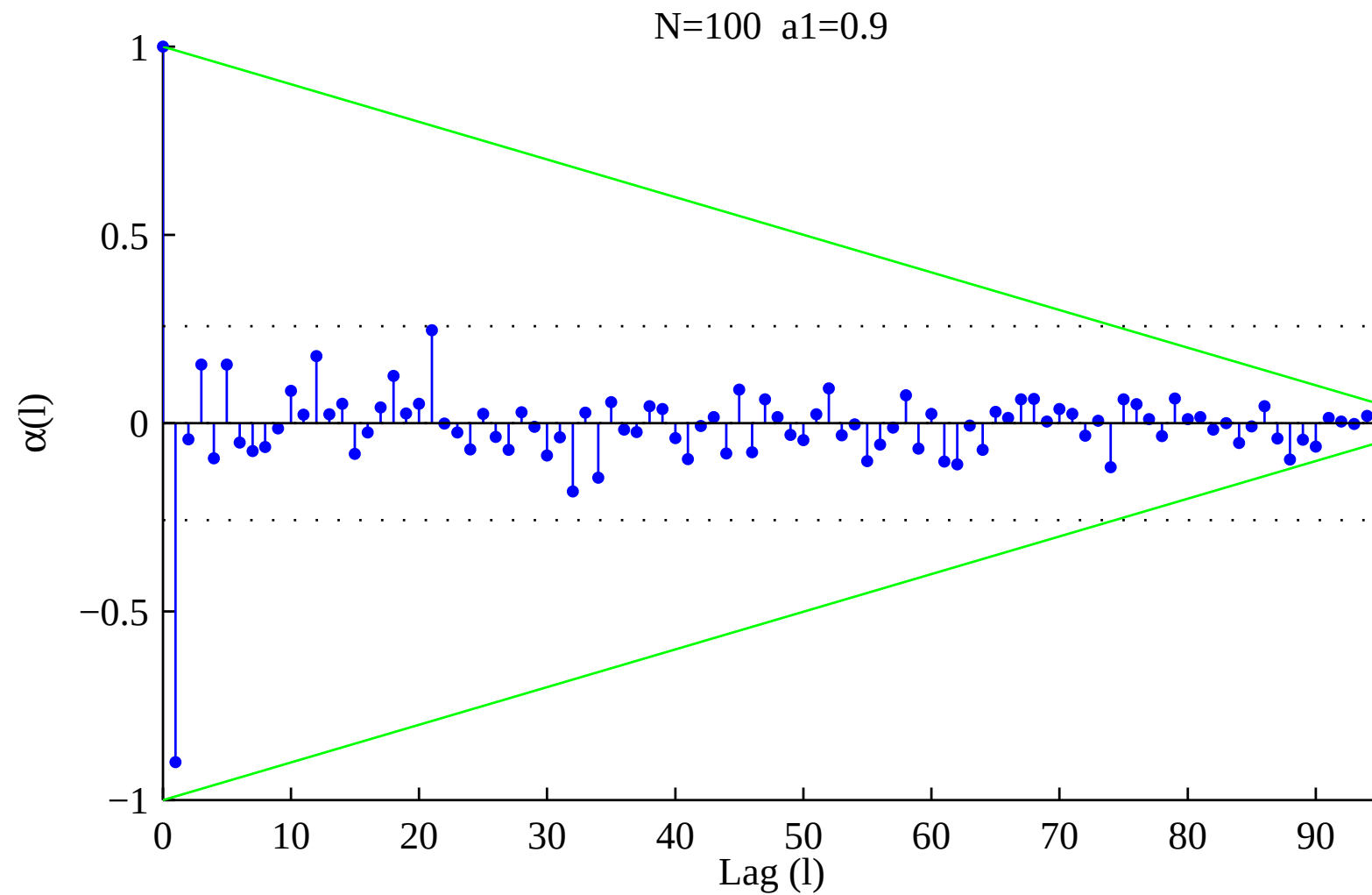
Example 5: AR(1) ACF



Example 5: AR(1) ACF



Example 5: AR(1) PACF



Example 5: MATLAB Code

```
%function [] = Autoregressive();
close all;
clear;

L = 95; % Length of autocorrelation calculated
A = [-0.95 0 0.5 0.95]; % Coefficients
sw = 1; % White noise power
ac = zeros(L+1,1);
N = 100;
cl = 99; % Confidence level

np = norminv((1-cl/100)/2); % Find corresponding lower percentile of Normal

%=====
% Main Loop
%=====
for c1 = 1:length(A),

    a1 = A(c1);

    ac(1) = sw/(1-a1^2);
    for c2=2:L+1,
        ac(c2) = -a1*ac(c2-1);
    end;
    acf = ac/ac(1);

    w = randn(N,1);
```

```

a = [1 a1];
x = filter(1,a,w);

%=====
% Plot the Signal
%=====
figure
FigureSet(1,'LTX');
n = 0:N-1;
h = stem(n,x);
set(h(1),'MarkerFaceColor','b');
set(h(1),'MarkerSize',2);
xlim([0 N]);
ylim([min(x) max(x)]);
ylabel('x(n)');
xlabel('Sample (n)');
title(sprintf('N=%d  a1=%3.1f',N,a1));
box off;
AxisLines;
AxisSet(8);
eval(sprintf('print -depsc AREstimateSignal%d;',c1));

%=====
% Plot the Biased ACF
%=====
[ac,n] = Autocorrelation(x,1,L+1,1); % Biased Estimate (BSP toolbox function)

figure
FigureSet(1,'LTX');
l = 0:L;
h = stem(l,ac);

```

```

set(h(1),'MarkerFaceColor','b');
set(h(1),'MarkerSize',2);
xlim([0 L]);
ylim([-1 1]);
hold on;
    plot(l,acf,'r. ');
    plot(xlim,np*[1 1]/sqrt(N),'k:',xlim,-np*[1 1]/sqrt(N),'k:');
    plot(xlim,[1 (N-L)/N],'g',xlim,-[1 (N-L)/N],'g');
    hold off;
ylabel('\rho(l)');
xlabel('Lag (l)');
title(sprintf('N=%d  a1=%3.1f',N,a1));
box off;
AxisLines;
AxisSet(8);
eval(sprintf('print -depsc AREstimateBiased%d;',c1));

%=====
% Plot the Unbiased ACF
%=====
[ac,n] = Autocorrelation(x,1,L+1,2); % Biased Estimate (BSP toolbox function)

figure
FigureSet(1,'LTX');
l = 0:L;
h = stem(l,ac);
set(h(1),'MarkerFaceColor','b');
set(h(1),'MarkerSize',2);
xlim([0 L]);
ylim([-1 1]);
hold on;

```

```

        plot(l,acf,'r. ');
        plot(l,np*sqrt(N)./(N-1),'k:',l,-np*sqrt(N)./(N-1),'k:');
        hold off;
ylabel('\rho(l)');
xlabel('Lag (l)');
title(sprintf('N=%d  a1=%3.1f',N,a1));
box off;
AxisLines;
AxisSet(8);
eval(sprintf('print -depsc AREstimateUnbiased%d;',c1));

%=====
% Plot the Biased PACF
%=====
[pac,n] = PartialAutocorrelation(x,1,L+1,2); % Biased Estimate (BSP toolbox function)

figure
FigureSet(1,'LTX');
l = 0:L;
h = stem(l,pac);
set(h(1),'MarkerFaceColor','b');
set(h(1),'MarkerSize',2);
xlim([0 L]);
ylim([-1 1]);
hold on;
        plot(xlim,np*[1 1]/sqrt(N),'k:',xlim,-np*[1 1]/sqrt(N),'k:');
        plot(xlim,[1 (N-L)/N],'g',xlim,-[1 (N-L)/N],'g');
        hold off;
ylabel('\alpha(l)');
xlabel('Lag (l)');
title(sprintf('N=%d  a1=%3.1f',N,a1));

```

```
box off;  
AxisLines;  
AxisSet(8);  
eval(sprintf('print -depsc AREstimatePACF%d;',c1));  
  
end;
```

Summary

- ACF and PACF are useful characterizations of WSS random processes
- Can help select an appropriate model
 - MA: Finite ACF
 - AR: Finite PACF
- AP/AR are often preferred characterizations because we can solve/estimate the model parameters by solving a set of linear equations (Yule-Walker)
- Biased estimates of $r(\ell)$, $\rho(\ell)$, and/or $\gamma(\ell)$ are generally preferred to the unbiased estimates
 - Less variance (always), and lower MSE (sometimes)
 - Positive definite (PSD is therefore also nonnegative)
- Bias is known, but variance of estimates is generally unknown
- Loosely, confidence intervals for WN are used instead